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Understanding the dynamic changes in wetland cultural ecosystem services: Integrating annual social media data into the SolVES

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ABSTRACT

As urbanization increases, the human demand for the cultural services of wetland ecosystems is growing. However, traditional cultural ecosystem service (CES) assessments are usually conducted using a focused survey over a short period, often ignoring the changes in CESs over time. Social media data have great potential for assessing dynamic changes in CESs. In this paper, we integrated social media data into the Social Value for Ecosystem Services (SolVES) model to identify and locate the CES values of Shanghai's Wusong Paotaiwan Wetland (WPW) Park in 2015, 2017, and 2019. We also quantified the inter-annual variation in the CES values through raster calculations. We found that the spatial distribution of the CES values showed a tendency towards mean values, which may be related to social media promotion, the improvements in park facilities, and visitors' play psychology of seeking differences. These results can help planners respond to visitor feedback and improve park programmes in a timely manner. Notably, the spatial distribution of the CES values varied significantly between years, which suggests that we should select data for different purposes and consider which year or years of data are most scientifically sound to facilitate reliable conclusions before conducting CES assessments.

1. Introduction

Wetlands are crucial to human survival and have the highest service value per unit area among all ecosystems (Gedan et al., 2011). However, as a result of both global warming and growing urbanization, wetlands have also suffered among the greatest environmental losses globally (Mondal et al., 2017) and are at risk of rapid degradation and disappearance. As an organic combination of wetland conservation, ecological restoration, and the sustainable use of wetland resources (Skeat, 1986), wetland parks have developed rapidly in China in the last decade, with over 800 new national wetland parks added, totalling more than 360,000 ha (Zhou et al., 2020). Wetland parks not only hold significance for maintaining natural functions such as urban health (Chaiyarat et al., 2019) and landscape diversity (Hartter & Southworth, 2009) but also embody the beautiful vision of providing multiple cultural ecosystem services (CESs), e.g., aesthetic and recreational services are the main CESs that wetland parks can offer (Pedersen et al., 2019). However, in the actual management and use of wetland parks, more attention is often paid to natural ecological services (Cohen-Shacham et al., 2015; Duan et al., 2017), with less focus on CESs.

The CESs of wetland parks often refer to the non-material advantages that people receive from visiting them, including spiritual uplift, cognitive growth, time for introspection, enjoyment, and sightseeing experience, and they involve a range of service types, such as aesthetic, recreational, therapeutic, historic, learning, and life-sustaining services (MEA, 2005). CESs are more immediately felt, understood, and experienced by humans than other ecosystem services, and they play an important role in increasing human well-being (Andersson et al., 2015). They are the key to attracting visitors to visit wetland parks and improving the competitiveness of such parks. Therefore, how to quantify and evaluate the values of wetland park CESs is crucial for ensuring the healthy and continuous development of wetland parks.

The two main categories of CES assessment methods are monetary and non-monetary approaches (Klain & Chan, 2012). Among the

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mainstream monetary approaches, the travel cost method (Van Berkel & Verburg, 2014) and the market price method (Zang et al., 2011) are more common. However, expressing CESs with multiple value attributes through economic means only is controversial (Liu et al., 2010), and apart from tourism and recreational services, other cultural services, such as aesthetic, educational, and historic services, are hardly reflected by economic indicators (Christie et al., 2012). If tourism and recreational services are used to represent CESs overall, then other CESs will be marginalized, and researchers and policymakers will be misled. With the deepening of CES-related research, researchers are increasingly using non-monetary methods to assess the value of CESs (Hirons et al., 2016). Although such methods cannot intuitively use data for monetization, they can explain the value of CESs in terms of importance classification (Czembrowski et al., 2016) and respondents' perceptions and preferences (Bryce et al., 2016; Schmidt et al., 2016). Additionally, mapping studies on CESs started to emerge in approximately 2011 as a result of the advances in computer technology and the iteration of relevant software updates (Sherrouse et al., 2011). The United States Geological Survey (USGS) created the Social Value for Ecosystem Services (SolVES) (http://solves.cr.usgs.gov), which allows a more comprehensive assessment, quantification, and mapping of CESs, such as recreational, aesthetic, historic, educational, and biodiversity perceptions (Sherrouse & Semmens, 2014). The maximum entropy (Maxent) modelling program, which is integrated into the SolVES, can produce statistical models to explain the connection between CES values and environment layers, complete the visual mapping of the assessment findings, and then create a composite report. The SolVES is considered a valuable model that can provide an important reference for planners to identify the CES value distribution of a region.

Even though there has been an increase in interest in CESs recently, the majority of research has been statically performed, typically using focused surveys for analysis over a short period (Plieninger et al., 2013). However, CESs are not static. According to some scholars (Raudsepp-Hearne et al., 2010; Wang et al., 2021), CESs evolve. However, the accuracy over time has rarely been discussed in previous studies. Therefore, we believe that assessments carried out for CESs should also be updated in a timely manner. At the same time, with the development of industrialization and urbanization, urban residents are increasingly distant from the natural environment (Thiagarajah et al., 2015), residents are increasingly eager to come into contact with nature, and their demand for CESs provided by wetland parks is also rapidly increasing (Martinico et al., 2014).

Against this backdrop, we collect data from multiple surveys in 2015, 2017, and 2019 and use the SolVES to assess and map the distribution of CES values, including aesthetic, historic, and recreational values. This paper focuses on Shanghai's Wusong Paotaiwan Wetland (WPW) Park as a study site, and it is the largest native wetland park in Shanghai and a reliable supplier of CESs. In addition, WPW Park is distinguished from other wetland parks by its many historical sites. We explore the dynamic changes in CES values over time by posing the following questions: (1) How and why do CES values alter over time? (2) What is the value of the results of the inter-annual variation in CES values to policymakers? (3) How can inter-annual data be reasonably combined to meet the needs of CES assessments for different purposes?

2. Method

2.1. Study area

The WPW Park is situated on a riverfront mudflat where the Yangtze and Huangpu Rivers meet in Shanghai, China. The area has been backfilled with steel slag since the 1960s and was named after the naval fortress built there by the Qing government. The park has a land area of 53.46 ha and a native wetland area of approximately 50 ha, with a shoreline of 1974.13 m. It is also currently the largest wetland park in Shanghai. The park was officially opened in 2007, and its design emphasizes the concept of cultural regeneration and ecological restoration. It is divided into four major regions: the forest recreation and sightseeing area, waterfront wetland landscape area, desert landscape area, and valley ecological corridor area. Additionally, it features attractions such as the Pit Garden, the Yangtze Estuary Science and Technology Museum, the Shell Theatre, the Marina, and the Sports Club, which can provide a broad variety of cultural services (see Fig. 1)

2.2. Survey data

The data for the study came from Meituan Dianping, a social media platform on which tourists can post their images and travel experiences. The photos uploaded by visitors contain information about a wide range of landscape features, such as forests, lakes, and wetlands, and they can help identify these locations and associated CES types. We queried the platform for visitors' comments and downloaded photos of the WPW Park in 2015, 2017, and 2019. After filtering out close-up photos that could not be linked to a specific location, we made several visits to the park with GPS locators and determined the latitude and longitude coordinates of all the photos. We found that aesthetic value, historic value, and recreational value are the main types of CES that the WPW Park can provide. Therefore, we classified the photos based on the descriptions of CESs (Table 1). Each CES value was given a score equal to 100 divided by the number of CES values in each review (Tian et al., 2021). In addition, photos of the interior were not used as valid data in this study.

2.3. Environmental GIS data

The SolVES requires a variety of geographical and tabular data to be entered into the ArcGIS personal Geodatabase, including three types. The first type is the tabular data that come with SolVES 3.0, including ATTITUDE_TYPES, USE_ATTITUDE, USE_TYPES, VALUE_ALLOCATION, and VALUE_TYPES, and we need to reload our survey data into these tables. The second type is the Shapefile, including SURVEY_POINT and STUDY_AREA. The former is based on survey data to determine and label the longitude and latitude coordinates, while the latter is based on remote sensing images for digitization. The third type is raster data; for this study, distance to roads (DTR), distance to wetlands (DTW), and distance to attractions (DTA) raster data were selected. This part of the data was taken from the map published on the park's official website, which was vectorized and then calculated using Euclidean distances included in the ArcGIS geographic analysis tool (see Table 2).

2.4. The SolVES model

The SolVES model consists of three sub-modules: the ecosystem services social-values module, the value mapping module, and the value transfer mapping module. The first two should be evaluated based on a large amount of survey data. The last module can transfer the output statistical model to a similar area without survey data to predict the value of ecosystem services in a similar area (Sherrouse & Semmens, 2014).

In this study, the ecosystem services social-values module and the value mapping module of the SolVES were applied. The SolVES model was run as follows: (1) The kernel density analysis tool integrated into the SolVES was used to conduct a weighted kernel density analysis of all CES value points taken by visitors. (2) The model performed mean nearest-neighbour analysis for all CES value types, and the spatial distribution patterns of CESs were judged by the mean nearest-neighbour ratio (R) and standard deviation (Z). (3) The model normalized the values assigned by CES types supported by visitors to obtain a 10-point value index (VI), and the highest VI among CES types was the maximum value index (M-VI) of this type. (4) The Maxent modelling software Maxent embedded in the SolVES was used to process the environmental data to produce the relationship between the CES values that were output in the previous step and the environment layers. (5) Using the

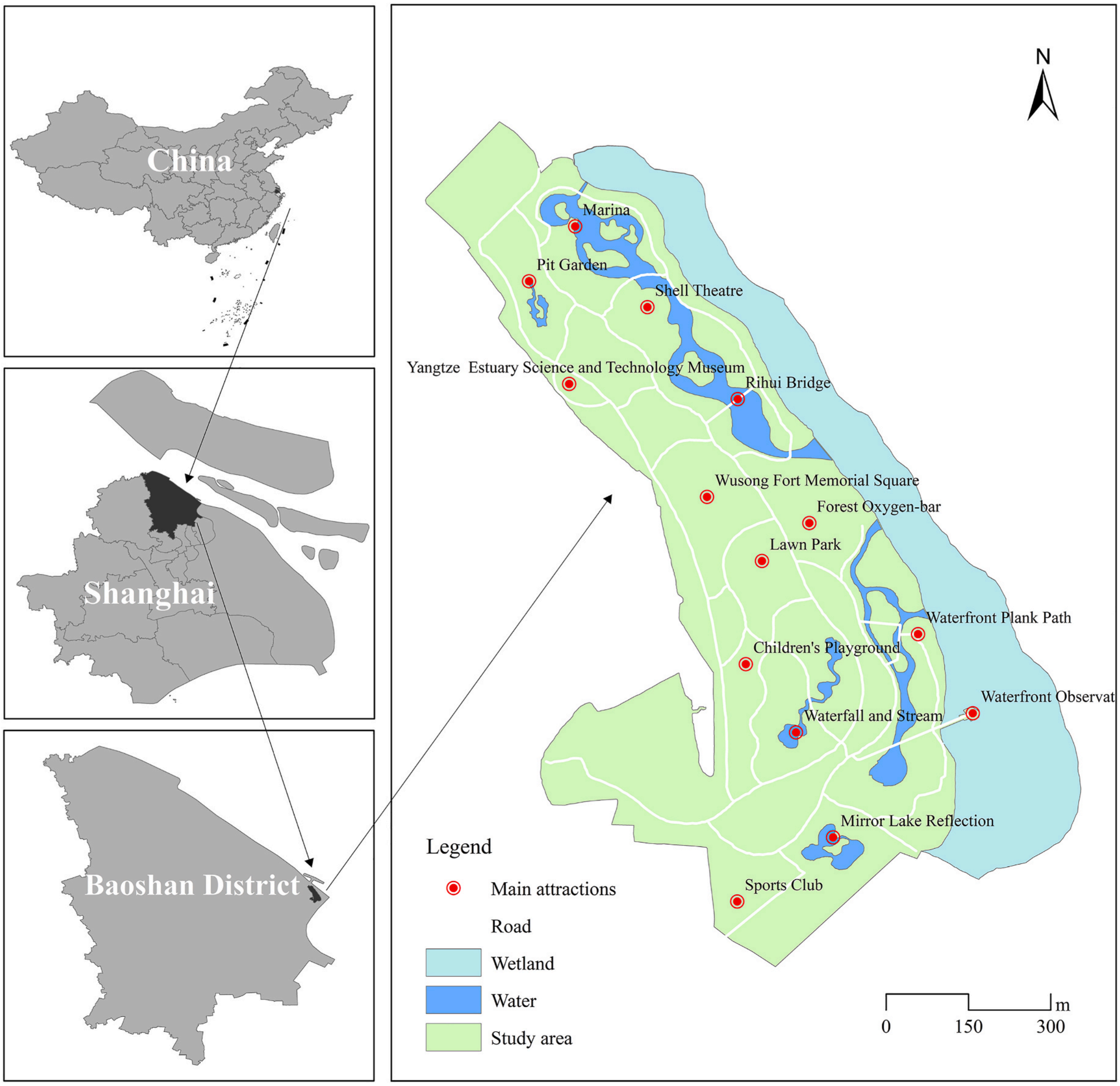

Fig. 1. Map displaying the location of the study area and the main attractions.

Table 1
Explanations and examples of CES value types used in this paper
Adapted from Wang et al. (2016).

CES value type	Explanation	Examples
Aesthetic value	I cherish this aspect because I love to see the landscape, hear the sounds of nature, and smell the fragrance of nature	● Seasonal and colour phases of plant landscapes ● Richness of the water body landscape ● Beautiful facilities and architecture
Historic value	I cherish this aspect because it is a historic heritage of both nature and people that is important to me, to other people, and to my nation	● Historic sites ● Local traditional cultural resources
Recreational value	I cherish this aspect because it offers a variety of places and opportunities for outdoor recreation	● Recreational activities in the environment (e.g. walking, crabbing, boating, fishing, playing) ● Artistic creation in the environment (e.g. painting, dancing, playing instruments, singing)

Table 2
Environmental GIS data to be used in SolVES.

Tabular data	Shapefile	Raster data
ATTITUDE_TYPES	SURVEY_POINT	Distance to roads (DTR)
USE_ATTITUDE		
USE_TYPES	STUDY_AREA	Distance to wetlands (DTW)
VALUE_ALLOCATION		
VALUE_TYPES		Distance to attractions (DTA)

value mapping module, a plot consisting of the corresponding kernel density surface, VI, and environmental data was derived to output a final spatial distribution subplot of the CES values.

3. Results

3.1. Location and density of CES points

Fig. 2 presents information on the location and density of CES points in 2015, 2017, and 2019. Over time, the proportion of grid cells with CES points in the total area increased significantly, and the maximum number of CES points held by a grid cell also increased, with 12, 95, and

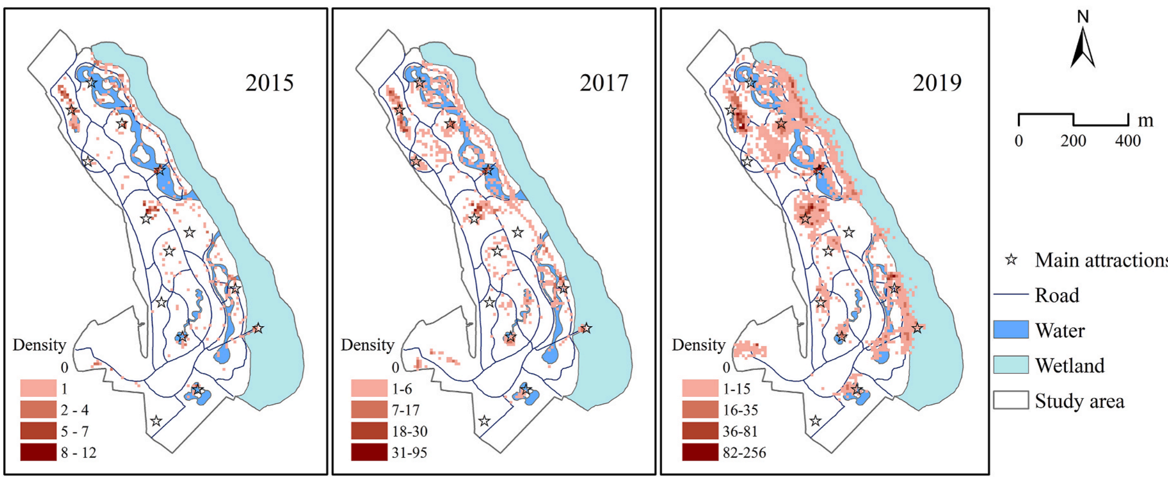

Fig. 2. Spatial distribution of CES points in different years at a spatial resolution of 10 m × 10 m.

256 for 2015, 2017, and 2019, respectively, indicating that the CES points were more closely distributed. Additionally, the inter-annual spatial distribution of CES points gradually tended to concentrate on the northern region of the park and the wetlands. In addition to the Wusong Fort Memorial Square and Pit Garden and other original popular attractions, there are some hydrophilic attractions, such as the Rihui Bridge and the Waterfront Plank Path, which gradually gained popularity among visitors.

3.2. M-VI and spatial distribution of CESs in different years

According to the findings of the mean nearest-neighbour analysis (Table 3), the spatial distribution of each value type in 2015, 2017, and 2019 all belonged to the aggregation pattern (the ratios were all less than 1, and the absolute values of the Z values were all larger) and are suitable for further analysis. The maximum value index for each CES is known as the M-VI, which can reflect the preference of visitors to a certain extent. The higher the value is, the stronger the preference. Comparing the M-VI of each CES in different years, we found that visitors prioritized aesthetic value the most. Except for 2015, the M-VI for aesthetic value was the same as that for historic value, and the order of importance for other years was aesthetic value > historic value > recreational value.

The VI map output from the SolVES can spatially reflect the distribution of the CES value index for 2015, 2017, and 2019 (Fig. 3). In general, the spatial distribution pattern of the CES VI shows a wider distribution range and a lower VI with inter-annual changes. Aesthetic value had the highest VI and the widest distribution, spreading over almost all areas of the park except for the interior of the wetlands and water area. Hotpots of aesthetic value were almost concentrated entirely in scenic attractions, such as the Mirror Lake Reflection, the Pit Garden, and the Waterfront Observation Deck. In contrast, the historic value was more sporadically distributed, with hotpots close to the Wusong Fort Memorial Square and the Yangtze River Estuary Science and Technology Museum. Recreational value was mainly concentrated along roads and in various attractions.

Table 3
M-VI and mean nearest-neighbour analysis for different years.

Year	CES value type	N_COUNT	R_RATIO	Z_SCORE	M-VI
2015	Aesthetic value	293	0.550469	−14.720555	9
	Historic value	69	0.208281	−12.581333	9
	Recreational value	98	0.539848	−8.71455	5
2017	Aesthetic value	1252	0.404663	−40.299137	9
	Historic value	310	0.162828	−28.198534	7
	Recreational value	445	0.424921	−23.208017	5
2019	Aesthetic value	6383	0.413211	−89.686197	7
	Historic value	1291	0.228833	−53.008129	3
	Recreational value	1818	0.358551	−52.3266	2

3.3. M-VI and spatial distribution of CESs in the three-year cumulative period

The result of the mean nearest-neighbour analysis (Table 4) show that the spatial distribution of each value type in the three-year cumulative period belonged to the aggregation pattern (the ratios were all less than 1, and the absolute values of the Z values were all larger), which is suitable for further analysis. The importance ranking based on the M-VI is aesthetic value > historic value > recreational value, which is the same as the ranking results for 2017 and 2019.

Fig. 4 reflects the spatial distribution of the three-year cumulative CES VI for each type. Compared with a single year, the overall distribution range of all kinds of CES VIs in the three-year cumulative period was wider, and the hotpots were roughly the same, but the area of the hotpots increased to different degrees, among which the area of aesthetic value hotpots increased the most obviously. The distribution of the three-year cumulative historic value and recreation value was more similar to that of 2019, with a certain expansion in the area with a high VI.

3.4. The distribution pattern of the inter-annual CES value change grade

To more intuitively demonstrate the distribution characteristics of the inter-annual variation in each type of CES value, the spatial raster data for each CES value every two years were subtracted to obtain the respective spatial distribution pattern change map and divided into five classes (Table 5) based on the corresponding degree of change: the strong decreasing area, weak decreasing area, unchanged area, weak increasing area, and strong increasing area (Fig. 5).

The inter-annual variation in aesthetic value was the most obvious (Table 6). In 2015–2017, the area mainly increased (10.09%), distributed in the central and western block areas of the park and the ring area near the wetland. The area of increase in 2017–2019 was smaller in size (6.58%), showing a clear distribution along the roads and spreading over almost all the sites in the park, while the area of decrease (9.08%) was concentrated in the southeast corner with fewer scenic spots and the northernmost part of the park. Notably, there was a strong increase in the area around the park gate in 2015–2017 and 2017–2019, even though there are no scenic spots.

The inter-annual variation in historic value was dominated by decreases. The area of decrease in 2015–2017 and 2017–2019 was 4.02% and 2.54%, respectively, mainly in the central part of the park and the

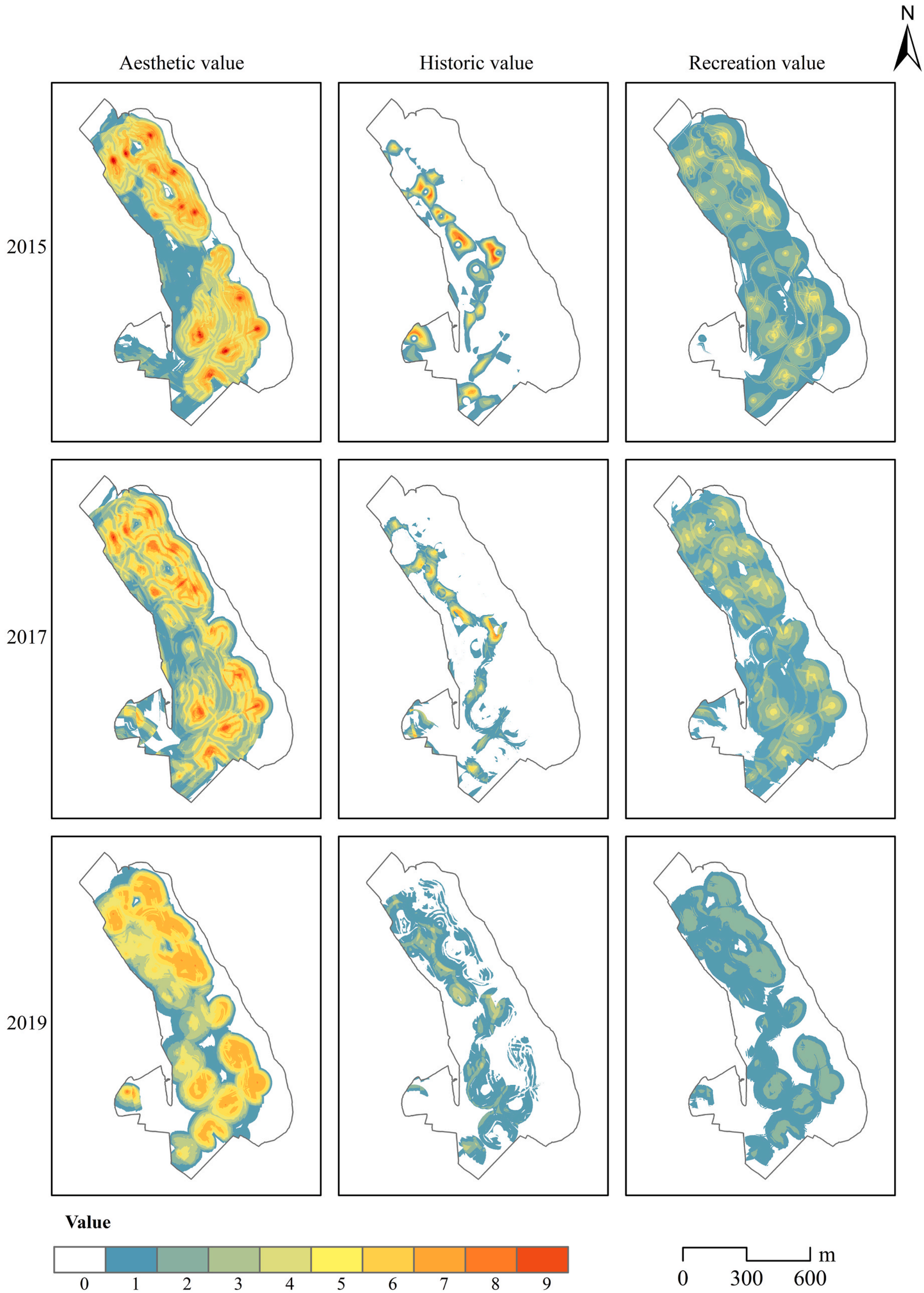

Fig. 3. Spatial distribution of the CES value index in different years: a) aesthetic value; b) historic value; c) recreational value.

Table 4
M-VI and the mean nearest-neighbour analysis in the three-year cumulative period.

CES value type	N_COUNT	R_RATIO	Z_SCORE	M-VI
Aesthetic value	7928	0.479972	−88.58095	9
Historic value	1670	0.224864	−60.599196	4
Recreational value	2364	0.435339	−52.522168	3

park gate, while the area of increase was no more than 1%, with an increase in the area of the Wusong Fort Memorial Square in 2017–2019.

The inter-annual variation in recreation value transitioned from relative stability to an overall decrease. A total of 98.60% of the unchanged area in 2015–2017 increased, and the decreasing area was sporadically distributed throughout the park. Additionally, 7.18% of the decreasing area in 2017–2019 was distributed mainly in the park's various attractions, with a greater concentration in the northern part of the park.

4. Discussion

4.1. The potential of using social media data in assessing CESs dynamics

The importance of assessing the dynamics of CESs has been widely recognized (Thiagarajah et al., 2015; Wang et al., 2021), but the practice of this recognition in the design and management of urban green is still relatively low. This low realisation is due in part to human investment and legal constraints that make it difficult for researchers to access data on public preferences over longer time scales (Tian et al., 2023). In recent years, social media use has become more widespread due to lower usage costs, increased network coverage, and improved connection speed (Chugh & Joshi, 2020; Schober et al., 2016). Social media-based methods offer free access to large amounts of data on public preferences, and social media platforms also tend to retain comments for many years. Therefore, a social media-based evaluation of CESs is a promising evaluation method (Guerrero et al., 2016).

But there is a problem with the application of social media-based approaches in that the number of unique users of social media always increases over time (Jay et al., 2022). For example, from 2015 to 2019, the cumulative number of independent users of the Meituan Dianping mobile client, which is the source of the data in this study, increased from 250 million to 400 million. Due to this direct impact, the number of CES points available for WPW Park increased from 460 to 9492. We argue that when using a social media approach as a data source, considering the evaluation of CESs from a quantitative perspective alone tends to yield convergent results. Therefore, this study provides a method for assessing CESs from a spatial display perspective to compensate for the use of social media as a data source. The Social Value for Ecosystem Services (SolVES) developed by the United States Geological Survey (USUG) was applied in this study, and this approach aims to assess the relative value of ecosystem services as perceived by different stakeholder groups and to generate maps of the value of ecosystem services (Sherrouse et al., 2017). Therefore, this study integrates social media data from 2015, 2017, and 2019 into the SolVES model, which can show changes in the relative distribution of the three CES types over three years in space.

4.2. Trends in inter-annual variation and influencing factors

We found that the spatial distribution of all types of CES values tended to be homogenized (i.e., the CES VI was lower and more widely distributed over time. The SolVES model generated weighted kernel density surfaces for all three CES types based on the points assigned to each CES type and scores. Points with higher amounts and the scores allocated to them receive a greater weighting and thus have a higher density of values (Sherrouse et al., 2011). Although the number of survey points was much higher in 2019 than in the other two years, their relative aggregation in space was not as high as that in 2015 and 2017. Therefore, it is clear that this change leads to a reduction in the maximum index identified by the model.

So what exactly causes the decrease in the relative spatial aggregation of CESs? We assumed the following scenario. At the beginning of the park's development, a small number of visitors tended to visit the most iconic spots in the park (e.g., the Wusong Fort Memorial Square and the Pit Garden). As the infrastructure and landscape of the park continue to improve, there are more places to play and enjoy the park, and the large number of visitors attracted by social media platforms have more choices to play. As the park's infrastructure and landscape have improved, the large number of visitors drawn by social media platforms have had more options to visit. At the same time, due to the differences in the preferences of each individual in a group (Ho et al., 2015),

Table 5
Classification of the degree of change in CES.

Strong decreasing area	Weak decreasing area	Unchanged area	Weak increasing area	Strong increasing area
[-6–3)	[-3-0)	0	(0–3]	(3–6]

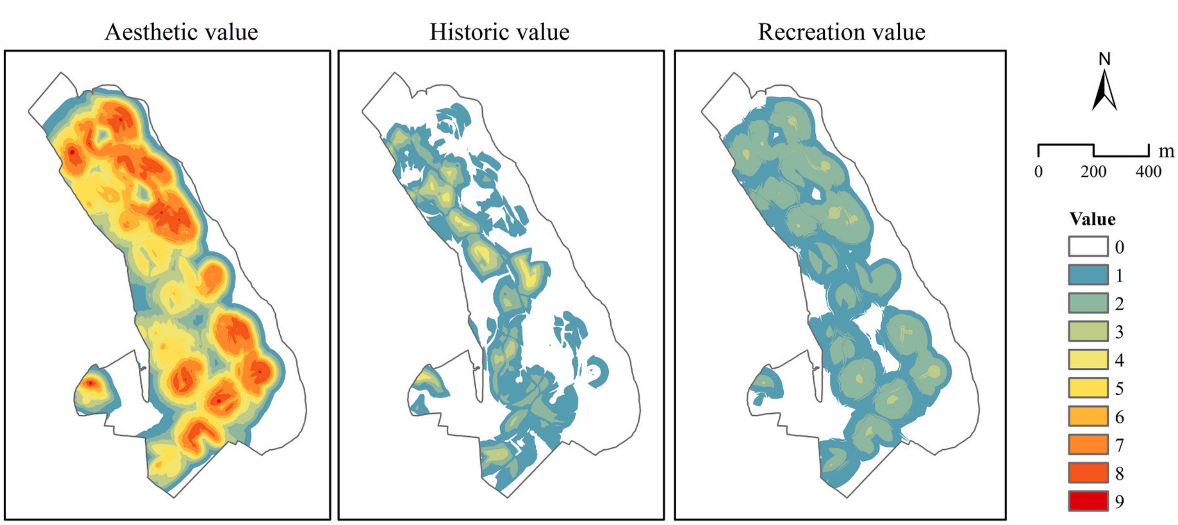

Fig. 4. Spatial distribution of the CES value index in the three-year cumulative period.

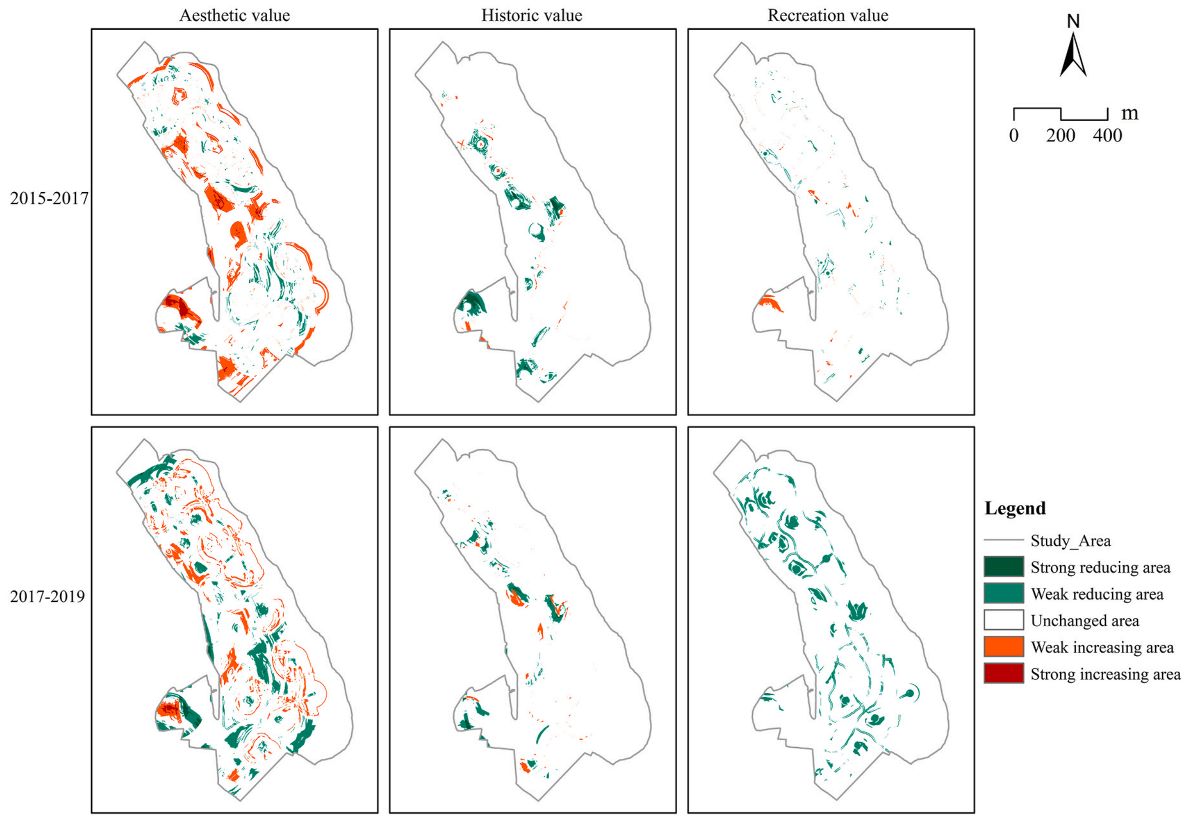

Fig. 5. Inter-annual distribution pattern of CES value index change classes.

Table 6
Area ratios for CES value index change classes.

Item	Strong decreasing area	Weak decreasing area	Unchanged area	Weak increasing area	Strong increasing area
Aesthetic value in 2015–2017	0.00%	2.75%	87.15%	9.26%	0.83%
Aesthetic value in 2017–2019	0.41%	8.67%	84.35%	6.40%	0.18%
Historic value in 2015–2017	0.65%	3.37%	95.54%	0.45%	0.00%
Historic value in 2017–2019	0.21%	2.33%	96.48%	0.98%	0.00%
Recreational value in 2015–2017	0.00%	0.72%	98.60%	0.68%	0.00%
Recreational value in 2017–2019	0.00%	7.18%	92.82%	0.00%	0.00%

visitors' behavior is generally more autonomous and spontaneous as the number of visitors increases, so it will show a more dispersed state in terms of the spatial distribution. In addition, for visitors interested in making multiple visits to the park, the psychological demand for novelty causes visitors to no be longer satisfied with repeating their past play activities (Mitas & Bastiaansen, 2018; Skavronskaya et al., 2020), and they are also more likely to choose scenic spots to explore. Previously, Scuderi and Dalle (2018) found that familiarity reduces visitors' landscape preference, which also supports this speculation. Therefore, from 2015 to 2019, the spatial distribution of each CES value showed an obvious diffusion trend, diverging and extending around the classic core scenic spots.

4.3. Value to policymakers

The SolVES provides a useful toolset that can be used to effectively capture and analyse visitors' visitation preferences for the WPW Park (van Riper et al., 2012). The output information of the aggregation degree and distribution of each CES value type can be used as a basis for park project improvement. In addition, the inter-annual variation in CES values can help policymakers make informed decisions in response to visitation feedback from visitors in a timely manner. For example, aesthetic value is the most widely distributed in the park, i.e., throughout the park except for the interior of the wetlands. Aesthetic value emphasizes the sense of beauty (Schirpke et al., 2016), and it often comes from the diversity of the landscape and broad panoramic views of forests, wetlands, meadows, etc. (Lee, 2017). Therefore, parks need to maintain the corresponding landscapes in the area where aesthetic value is distributed and pay focused attention to the decreasing area for the continuous quality of aesthetic landscapes. The distribution of historic value is scattered, and the inter-annual variation is mainly reduced, which may be a common pain point in China's public space (Zhou et al., 2015). Compared with other values, the historic value may not be easily perceived by visitors. Indeed, historic services may be the most valuable contribution made by urban parks (Gabdrakhmanov, 2016; Van Berkel et al., 2018). Historic value reminds us of our collective and individual roots (Daw et al., 2016), as well as the inheritance of shared human memories and values. Although the park has built attractions with historic and educational functions (e.g., the Wusong Fort Memorial Square, and Yangtze River Estuary Science, and Technology Museum), they are not well known to visitors, and it is still necessary to further grow their historic value. In addition, it is necessary to expand the publicity of these scenic spots through multiple platforms, both online and offline, to improve their attractiveness to visitors. The VI for recreational value is low and continues to decrease over the years. Compared with the other two values, the competitiveness of recreational value is weak, indicating that there are fewer experiential play items for visitors in the park and more of them are sightseeing tours. Both the growth of tourist initiatives and the building of infrastructure should receive further support from policymakers, who should also place a high value on visitor satisfaction.

4.4. Adaptation of data sets

The SolVES was originally developed by Sherrouse et al. (2011). They used examples from the Pike and San Isabel National Forests to show how the SolVES might be used to analyse, map, and quantify CESs. This was the beginning of the use of SolVES. Since then, many scholars have extended the SolVES to evaluate CESs from several perspectives. For example, the data sources are not limited to the traditional questionnaire method, and visitor-employed photography (VEP) (Sun et al., 2019) and the social media method (Tian et al., 2021) have become important ways to collect data for the SolVES in recent years. In addition, the application of SolVES has expanded from national forests (Sherrouse & Semmens, 2014) to agricultural land (Makovníková et al., 2016), landscape reserves (Zhao et al., 2019), wetland parks (Zhou et al., 2020), etc. The research of these scholars provides a valuable reference for our study. However, they mostly choose data from a particular year or a single survey in the selection of data, and using the SolVES to assess CESs at different time scales needs to be further explored. We find that there are significant differences between years in the spatial distribution of each CES value type (Fig. 3), and there may be a large bias if data from only one year or one survey are used in the actual evaluation. Therefore, we propose the idea that data should be combined based on different purposes before carrying out CES value assessment. For example, the comprehensive situation of a park can be more scientifically and reasonably reflected by assessing data from many years or multiple surveys (Fig. 4), while it is more appropriate to use data from the latest survey or the most recent year for the design improvement of a park. The combined analysis of different data sets can make the assessment results more effective and more targeted.

4.5. Limitations of the study and future prospects

Although our study was carefully designed, there are still some limitations that warrant further research attention.

First, we selected only three years of data to study the inter-annual variation in CES values, and there may be some contingency in the presentation of the results. For accurate extrapolation, further improvements to the research design are needed, such as adding offline interviews with visitors as supplementary research, summarizing and introducing possible parameters, and incorporating longer time series for quantitative analysis to obtain more reliable conclusions.

Second, there may be some labelling bias when using social media as a data source to analyse CES values. For example, we find that both the aesthetic and recreational values near the gate of the park increased over two years. However, it is hard to discern whether people just use the gate as a geographical reference or discuss it on social media because it is attractive. In the future, a more robust unsupervised selection method will be required to assess these data sets and enhance the precision of the outcomes.

5. Conclusion

This study used the SolVES model, combined with social media data, to assess and quantify the relative spatial variation in the CES value of the WPW parks in different years. We find that the inter-annual variation in the spatial distribution of each CES value shows the characteristics of diverging and extending from the classic core scenic spots to the surrounding area. Social media promotion, improvements in park facilities, and visitors' desire for the difference may be factors influencing the above changes in CES values. Planners should continue to maintain appropriate landscapes and facilities in the area where CES values are distributed and focus on the area where they are reduced. Due to the significant inter-annual variation in CES values, we believe it is scientifically reasonable to use multi-year or multiple survey data to assess CES values to provide an integrated park response, while it is more appropriate to use the latest survey or the most recent year for park design improvements.

CRediT authorship contribution statement

Sitong Huang: Conceptualization, Methodology, Writing – original draft. **Tian Tian:** Investigation, Data curation. **Lingge Zhai:** Software, Formal analysis. **Lingzhi Deng:** Validation, Project administration. **Yue Che:** Conceptualization, Supervision, Funding acquisition, Writing – review & editing.

Declarations of interest

None.

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